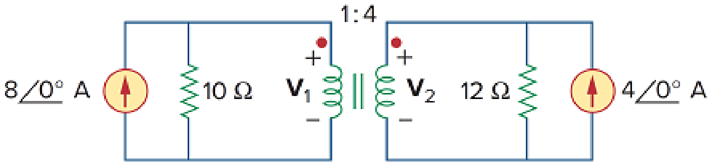
Chapter 13, Problem 43P

Problem

Obtain **V**1 and **V**2 in the ideal transformer circuit of Fig. 13.108.

Figure 13.108



Step-by-step solution

Step 1 of 6

Refer to Figure 13.108 in the textbook for the ideal transformer circuit.

Convert the current sources in the circuit into voltage sources.

Write the current at the input side of the transformer.

$$I_{s1} = 8\angle 0^\circ \text{ A}$$

Calculate the voltage at the input side of the transformer.

$$\begin{aligned} V_{s1} &= I_{s1}R \\ &= (8\angle 0^\circ \text{ A})(10\ \Omega) \\ &= 80\angle 0^\circ \text{ V} \end{aligned}$$

Write the current at the output side of the transformer.

$$I_{s2} = 4\angle 0^\circ \text{ A}$$

Calculate the voltage at the output side of the transformer.

$$\begin{aligned} V_{s2} &= I_{s2}R \\ &= (4\angle 0^\circ \text{ A})(12\ \Omega) \\ &= 48\angle 0^\circ \text{ V} \end{aligned}$$

Step 2 of 6

Draw the modified circuit as shown in Figure 1.

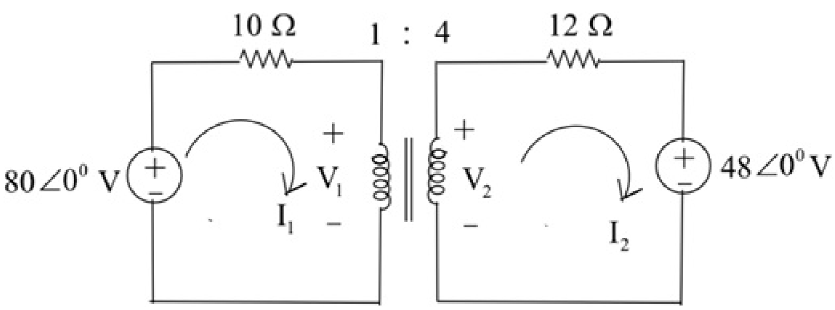


Figure 1

Step 3 of 6

Apply Kirchhoff's voltage law to the input mesh in the circuit.

$$10\mathbf{I}_1 + \mathbf{V}_1 - 80 = 0$$
$$10\mathbf{I}_1 + \mathbf{V}_1 = 80$$

Apply Kirchhoff's voltage law to the output mesh in the circuit.

$$12\mathbf{I}_2 + 48 - \mathbf{V}_2 = 0$$
$$12\mathbf{I}_2 - \mathbf{V}_2 = -48$$

Consider the following expression for the voltage ratio of the transformer:

$$\frac{\mathbf{V}_2}{\mathbf{V}_1} = \frac{N_2}{N_1}$$

Substitute 4 for N_2 and 1 for N_1.

$$\frac{\mathbf{V}_2}{\mathbf{V}_1} = \frac{4}{1}$$
$$\mathbf{V}_2 = 4\mathbf{V}_1$$

Consider the following expression for the current ratio of the transformer:

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{N_2}{N_1}$$

Substitute 4 for N_2 and 1 for N_1.

$$\frac{\mathbf{I}_1}{\mathbf{I}_2} = \frac{4}{1}$$
$$\mathbf{I}_2 = \frac{\mathbf{I}_1}{4}$$

Step 4 of 6

Substitute the expressions of $\mathbf{V}_2$ and $\mathbf{I}_2$ in the second KVL equation.

$$12\mathbf{I}_2 - \mathbf{V}_2 = -48$$
$$12\left(\frac{\mathbf{I}_1}{4}\right) - 4\mathbf{V}_1 = -48$$
$$3\mathbf{I}_1 - 4\mathbf{V}_1 = -48$$
$$3\mathbf{I}_1 = 4\mathbf{V}_1 - 48$$
$$\begin{aligned}\mathbf{I}_1 &= \frac{4}{3}\mathbf{V}_1 - \frac{48}{3} \\ &= 1.34\mathbf{V}_1 - 16\end{aligned}$$

Substitute $1.34\mathbf{V}_1 - 16$ for $\mathbf{I}_1$ in the first KVL equation.

$$10(1.34\mathbf{V}_1 - 16) + \mathbf{V}_1 = 80$$
$$13.34\mathbf{V}_1 - 160 + \mathbf{V}_1 = 80$$
$$14.34\mathbf{V}_1 = 80 + 160$$
$$\begin{aligned}\mathbf{V}_1 &= \frac{240}{14.34} \\ &= 16.75\ \text{V}\end{aligned}$$

Step 5 of 6

Substitute 16.75 V for $\mathbf{V}_1$ in the first KVL equation.

$$10\mathbf{I}_1 + \mathbf{V}_1 = 80$$
$$10\mathbf{I}_1 + 16.75 = 80$$
$$10\mathbf{I}_1 = 80 - 16.75$$
$$\mathbf{I}_1 = \frac{63.25}{10} = 6.325 \text{ A}$$

Step 6 of 6

Calculate the value of $\mathbf{I}_2$.

$$\mathbf{I}_2 = \frac{\mathbf{I}_1}{4} = \frac{6.325}{4} = 1.58 \text{ A}$$

Substitute 1.58 A for $\mathbf{I}_2$ in the second KVL equation.

$$12\mathbf{I}_2 - \mathbf{V}_2 = -48$$
$$12(1.58) - \mathbf{V}_2 = -48$$
$$18.976 - \mathbf{V}_2 = -48$$
$$\mathbf{V}_2 = 18.976 + 48 = 66.976 \text{ V}$$

Therefore, the values of $\mathbf{V}_1$ and $\mathbf{V}_2$ are $\boxed{16.75 \text{ V}}$ and $\boxed{66.976 \text{ V}}$ respectively.